

(A copy of 'Questionnaire' section's answer)

Dear Dr. Zarkali and Weillab's team,

I am writing to express my interest in the Research Assistant position focused on the characterisation of Lewy Body Dementia (LBD). My academic and research background has deeply fueled my passion for neuroscience. During my master's degree, I specialised in neuroscience and applied technologies, working on brain organoid models derived from induced pluripotent stem cells (iPSCs) with a focus on their potential application in neurodegenerative disease research. This work provided me with valuable insights into the complexities of accurately characterising and diagnosing such conditions. My experience in multi-disciplinary lab environments has equipped me to contribute effectively to research development, while maintaining a strong sense of responsibility in both laboratory and clinical settings.

During my undergraduate studies in Physics and Medical Physics, I built a solid foundation in theoretical concepts related to multi-modal medical imaging techniques, including MRI. Through courses in image processing, data analysis, and programming, I developed a strong understanding of the methodologies for collecting, processing, and analysing imaging data. My practical skills in tackling neurophysiological recordings and brain imaging data were further honed during my master's studies in neurotechnology and various research projects. For instance, in a Brain-Machine Interface project, I decoded monkey movement trajectories from single and multiple unit recordings, enhancing my ability to process large neural datasets and extract meaningful features using statistical and programming techniques. Additionally, I refined my programming skills, particularly in Python, MATLAB and C++, through applications in engineering design and medical statistics involving ultrasound and advanced microscopy. I believe these skills will be a significant advantage in the data-related work required in this position.

My research skills, developed across diverse lab environments, are highly transferable to the organisational and management tasks required for this role. In my previous role as a research assistant, I conducted pre-clinical characterisation of an ultrasound system with a newly launched GUI. This involved designing experimental procedures, calibrating software parameters, and drafting user guidance for a multi-disciplinary audience. This role equipped me with practical skills in data management, project development, and troubleshooting while enhancing my ability to communicate effectively with colleagues and external collaborators. Additionally, in my master's project focused on Alzheimer's disease, I cultured cerebral tissues derived from human and mouse iPSCs to study early-stage neuronal development. The wet lab work required precise sample management, experiments scheduling, and strict protocol adherence, significantly improving my project management skills and methodical approach. My engineering experience further includes building a prototype to translate

EMG signals into audio signals for muscle control biofeedback and designing components and electrical circuits for enhanced data acquisition. These experiences have greatly strengthened my problem-solving abilities, and reinforced commitment to delivering high-quality work through personal dedication and thoughtful collaboration.

The clinical work in this role would also demand strong interpersonal skills, which I have developed through diverse working and volunteering experiences. At a care house for elderly individuals, I observed that many residents had neurodegenerative diseases that significantly impacted their quality of life. In response, our student group organised handcrafting activities for the residents and launched campaigns to raise public awareness of neurodegenerative diseases. Additionally, my experience working with primary school student with learning difficulties refined my ability to communicate interactively, emphasising the importance of active listening and respecting individual paces. My part-time work in various sectors provided me with experience in building an inclusive, respectful environment for both customers and the team members, fostering engaging communication and collaboration. I consistently maintain confidentiality in all aspects of my work and am highly motivated to develop my abilities in participant-focused clinical research, particularly in neuropsychological testing.

My passion for understanding the brain spans from its underlying biological mechanisms, at the molecular level to behavioural and higher-level cognitive functions. This aligns closely with the research in your lab. The challenge of decoding the genetic, environmental and developmental factors that influence disease progression is a complex yet vital area of research. Identifying biomarkers is crucial for enabling accurate diagnosis and effective treatment. I am currently preparing to apply for a Ph.D. program, and I have a strong interest in the multidisciplinary studies at UCL Queen Square. Your esteemed team's research on the effects of Parkinson's disease on cognitive functions is particularly compelling to me, as my interest in neuroscience began with a project focused on understanding dysfunctional motor functions by analysing EMG signals that reflect neuromuscular interactions.

The opportunity to join your dynamic research team would be invaluable for advancing my research skills and preparing me for a Ph.D. I am eager to contribute to and learn from your team, and I am excited about the potential growth and development this position offers.

I look forward to hearing from you and discussing in more detail how my research interests and background align with this role.

Yours sincerely,

Chuqi